Non-profit finances and macroeconomic trends

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Economics Honors Thesis

What financial practices are most correlated with counter cyclicality in US 501(c)(3) non-profit organizations?

- Demand for non-profit goods & services (i.e., food banks or job training) increases in recessions, yet this demand is not always met.[1]
- Supply for non-profit funding (i.e., private donations or government grants) decreases in recessions.[2]
- Despite this, some non-profit organizations are more economically resilient than others, depending on financial practices.[3]

1. Exley, Christine L., Nils H. Lehr, and Stephen J. Terry. "Nonprofits in good times and bad times." *Journal of Political Economy Microeconomics* 1, no. 1 (2023): 42-79.
2. Tomasko, Laura, Hannah Martin, Katie Fallon, Mirae Kim, Lewis Faulk, and Elizabeth T. Boris. *How Government Funding Disruptions Affected Nonprofits in Early 2025: Nationally Representative Findings from the Nonprofit Trends and Impacts Study*. Urban Institute.
3. Arbogust, MacKenzie. "Why do nonprofits fail? A quantitative study of form 990 information in the years preceding closure." PhD diss., Old Dominion University, 2020.

Demand Side Regressions

- $Y_{program\ expenses} = \beta_{GDP} + \alpha_{fixed\ effects} + \mu$
- Do non-profits expand during a recession?
- (Looking $\beta < 0$ … $1\%\ increase\ in\ GDP\ decreases\ programming\ and\ vice\ versa$)

Looking for non-profits that expand to increased demand of charity

Supply Side Regressions

- $Y_{donation\ revenue} = \beta_{GDP} + \alpha_{fixed\ effects} + \mu$
- Do donations increase during a recession?
- (Looking $\beta < 0$ … $1\%\ decrease\ in\ GDP\ increases\ donations\ and\ vice\ versa$)

Looking for the *magical* dependent variable that enables expansion

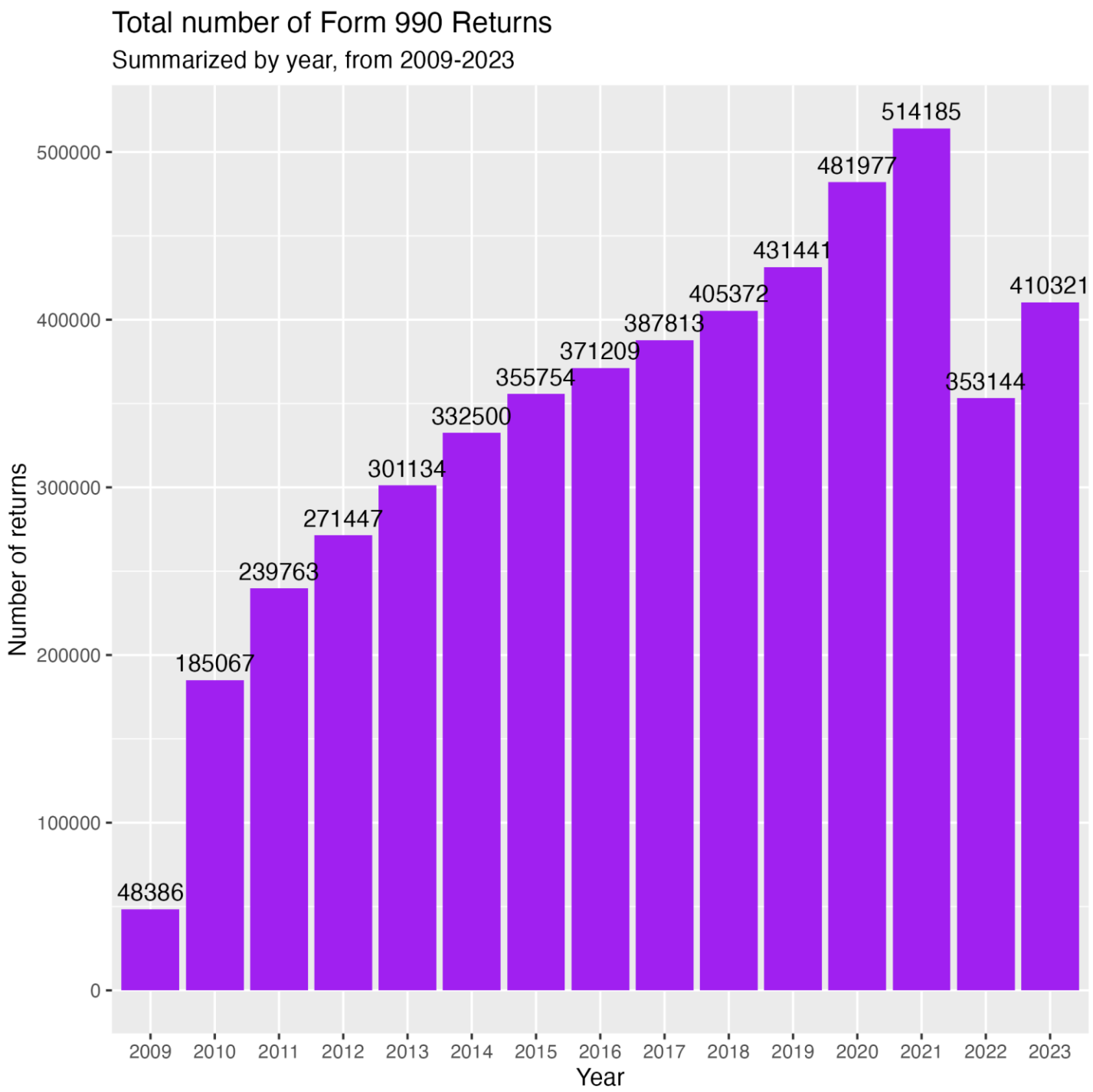
Revenue-controlled Regression

- $Y_{program\ expenses} = \beta_{GDP} + \beta_{revenue} + \alpha_{fixed\ effects} + \mu$
- Do non-profits still expand during a recession even if revenues are constant?
- (Looking $\beta < 0$ … $1\%\ decrease\ in\ GDP\ increases\ programming\ and\ vice\ versa$)

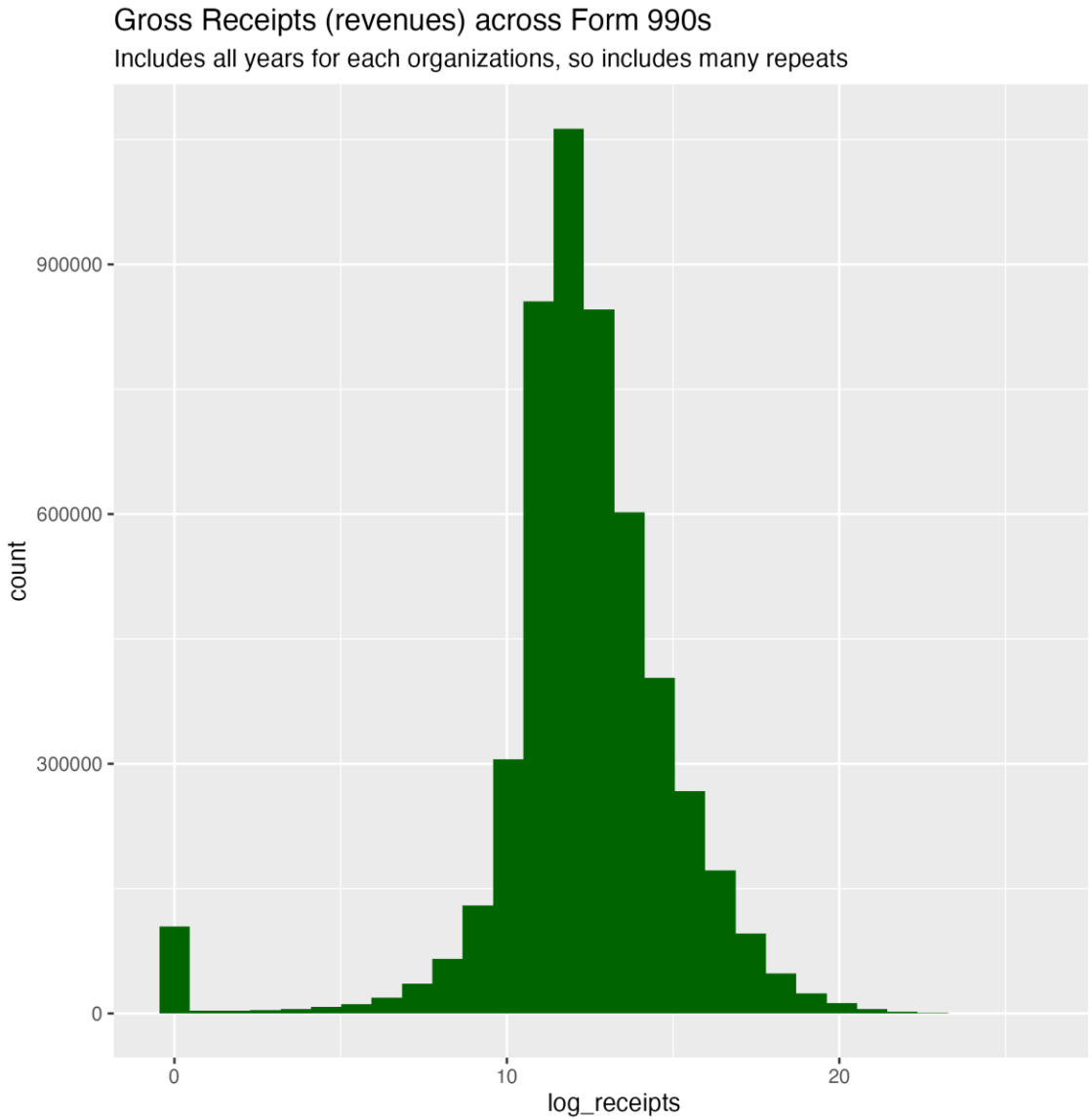
Verifying that non-profits expand due to *mission*, not *budget*

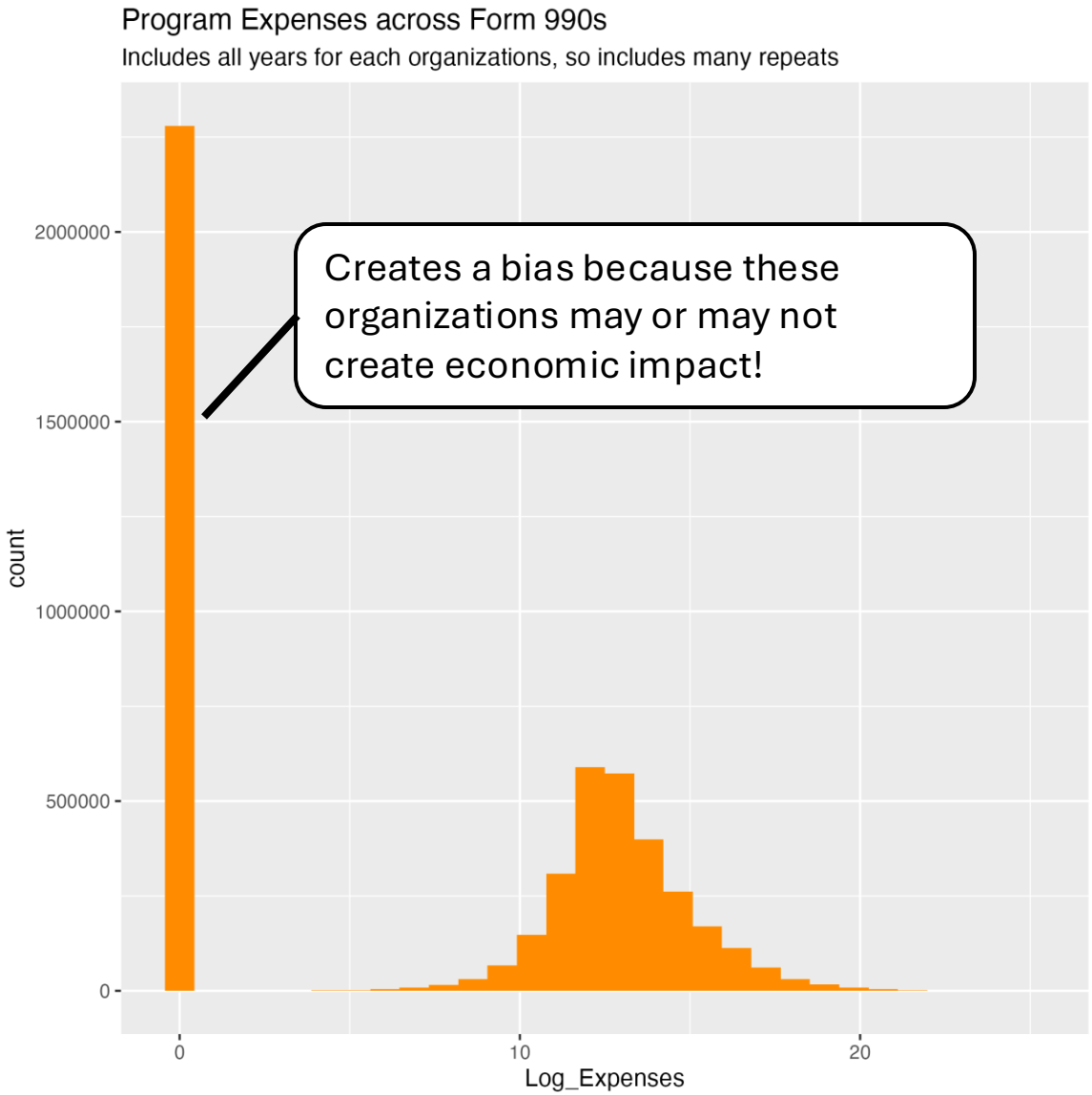
Meta Data

- 748,630 different non-profit organizations
- Approximately 5 million observations
- 15 years of completed tax returns
- Includes all 50 states + approximately 270 other U.S. territories or foreign states
- Includes all organizations / businesses that have gross receipts over $200,000 within the United States AND have filed for this income to be exempt from taxes
- Includes Clark University



Spread of Revenues / Expenses, by Size





lm(formula = Log_Expenses ~ GDP%, data = demand_side)

```
Residuals:
   Min     1Q Median     3Q    Max
-7.503 -7.178  3.528  5.826 17.927
Coefficients:
            Estimate Std. Error t value          Pr(>|t|)
(Intercept)  7.340651   0.004663 1574.14 <0.0000000000000002 ***
GDP         -0.063078   0.001534  -41.11 <0.0000000000000002 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 6.651 on 5089511 degrees of freedom
Multiple R-squared:  0.0003319,     Adjusted R-squared:  0.0003317
F-statistic:  1690 on 1 and 5089511 DF,  p-value: < 0.00000000000000022
```

lm(formula = Log_Fundr_Revenues ~ GDP%, data = supply_side)

```
Residuals:
   Min     1Q Median     3Q    Max
-1.047 -1.041 -1.040 -1.038 18.965
Coefficients:
            Estimate Std. Error t value            Pr(>|t|)
(Intercept) 1.0348767  0.0022493 460.089 < 0.0000000000000002 ***
GDP         0.0019818  0.0007401   2.678             0.00741 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 3.208 on 5089511 degrees of freedom
Multiple R-squared:  1.409e-06,	Adjusted R-squared:  1.212e-06
F-statistic:  7.17 on 1 and 5089511 DF,  p-value: 0.007415
```

lm(formula = Log_Expenses ~ GDP + Log_Revenues, data = revenue_controlled)

```
Residuals:
    Min      1Q   Median      3Q     Max
-20.8684  -0.0594   0.9595   1.8795  19.3423
Coefficients:
              Estimate Std. Error t value            Pr(>|t|)
(Intercept)   0.0752726  0.0032956   22.84 < 0.0000000000000002 ***
GDP          -0.0063047  0.0008373   -7.53   0.0000000000000508 ***
Log_Revenues  0.8419577  0.0002426 3469.92 < 0.0000000000000002 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 3.623 on 5074756 degrees of freedom
  (14754 observations deleted due to missingness)
Multiple R-squared:  0.7036,  Adjusted R-squared:  0.7036
F-statistic: 6.023e+06 on 2 and 5074756 DF,  p-value: < 0.00000000000000022
```

Next Steps

To improve my Regression analyses

- Control for state fixed-effects by including state specific GDP growth and non-profit headquarters
- Control for certain sectors of non-profits (i.e. school, hospital, food bank)
- Control for the size and age of the organizations

Why?

- Different states may be impacted in different magnitudes; this would help isolate the variance
- Non-profit actors may respond differently in the same economic conditions
- More experienced orgs. may be more adaptable